

Rajkumar Kuppuswami

Data Scientist/Data Engineering Specialist/AI & ML/Application Engineer/Business Intelligence

Data Engineering & Machine Learning with 13+ years of experience building high-impact AI/ML systems and data platforms at enterprise scale. Specializes in developing end-to-end ML solutions, from robust data pipelines to production model deployment and monitoring. Expertise in Python, SQL, and modern AI frameworks (PyTorch, TensorFlow), with a proven ability to deliver innovative solutions in Generative AI and predictive analytics. **Specialties:** End-to-End Machine Learning, Data Lake Architecture, Business Intelligence & Visualization, Supply Chain & Global Trade Analytics.

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EDUCATION

Master of Science in Data Science
Bellevue University, USA, Nebraska

07/2019 - 01/2022

Bachelor of Computer Science & Engineering
Anna University, India, Chennai

06/2006 - 05/2010

SKILLS

Domain & Functional Expertise	Semiconductor Manufacturing & Tool Processes, Supply Chain Analytics, Inventory Management, Sales Analytics, Global Trade & Logistics
Data Engineering & Big Data	Pandas, NumPy, Spark, Hadoop, SQL, PostgreSQL, Docker, Kubernetes, Airflow, ETL/ELT Pipelines, Data Modeling,
Cloud Data Platforms	Databricks, AWS, Azure, GIT
Programming & Scripting	Python, SQL, R, TensorFlow, PyTorch, scikit-learn, Neural Networks, NLP, LLM
Business Intelligence	ETL/ELT, Tableau, Power BI, Tableau Prep, Power BI, Grafana, Data Visualization & Storytelling, KPI & Executive Dashboarding
Machine Learning & Artificial Intelligence	Predictive Modeling, Supervised & Unsupervised Learning, Classification, Regression, Clustering Model Evaluation & Feature Engineering, Analytics
Statistical & Analytical Methods	Descriptive & Inferential Statistics, Hypothesis Testing, Regression Analysis, Time Series Analysis, Anomaly Detection
Databases & Storage	Relational Databases (MySQL, PostgreSQL, SQL Server), NoSQL Databases, Data Warehousing
Application Engineering	Software Development, Process Automation, API Integration, Troubleshooting & Root Cause Analysis
Tools & Platforms	Git/Version Control, Jupyter Notebook, RStudio, SharePoint, SAP ECC/SAP GTs

AEAR OF EXCELLENCE

- Data Engineering & Data Lake Architecture
- Big Data Platforms & Cloud Analytics
- Prompt Engineering/Vibe Coding
- Business Intelligence & Data Visualization
- Actionable Insights
- Project Mangement
- Process Improvement
- Claud
- Supply Chain, Global Trade & Logistics Analytics
- Statistical Analysis & Data Science
- AI/ML Algorithm Optimization
- Artificial Intelligence/Machine Learning
- Deep Learning & Predictive Modeling
- Cross-Functional Collaboration & Stakeholder Engagement
- Semiconductor Analytics & Tool Processes
- Automation & Application Engineering



WORK EXPERIENCE

Data Scientist

Applied Materials USA, Austin

09/2019 - Present

Austin

Key Projects & Business Impact

- **Demand—Supply Optimization Tool:** The team developed an end-to-end demand—supply analytics solution that integrated sales forecasts, inventory signals, and operational constraints. The tool enabled proactive planning decisions, reducing demand-supply mismatch and shortening planning cycles by approximately 40%. Performed Data Engineering using PySpark and built a dashboard using Tableau
- **Global Trade Revenue Prediction Model** I developed an ensemble machine learning model in Python, which achieved an accuracy rate of 92%. I developed the Flask API with Airflow automation, resulting in a reduction of forecast time from weeks to hours.
- **Tariff Impact Analytics Platform** Designed and deployed tariff impact Executive dashboards across the company to quantify duty exposure and cost implications across regions and product lines. The platform empowered leadership with near real-time insights, reducing **manual tariff analysis effort by over 60%** and enabling faster cost-mitigation strategies.
- **Tariff Modular — CBP Actuals Tracking** Built a scalable tariff module integrating **U.S. CBP import data** to track actual import values, duty payments, and variances against forecasts. Enhanced audit readiness and compliance reporting, **reducing reconciliation cycle time from multiple days to hours**.
- **Ethics & Compliance Fraud Detection (Concur Data)** Developed a **rule-based fraud detection framework** using Concur expense data to support investigations and ongoing monitoring. Automated compliance checks replaced manual reviews, **reducing investigation cycle time by ~70% and** strengthening preventive controls for ethics & compliance teams.
- **Python Automation:** Developed 20+ automation scripts (Pandas) for data extraction and RESTful APIs. Reduced manual effort by 50-60%, cutting operational costs 10%.
- **Time Series Supplier Analytics:** Built a predictive model (Python) forecasting OTD rates. Deployed on Databricks ML Framework with batch inference, improving supplier performance from 78% to 91%.
- Implemented advanced and innovative analytical techniques, algorithms, and tools to make data actionable and relevant to stakeholders through exploratory analysis of internal and external data sources
- Built a **centralized Purchasing Workbench** as a **one-stop analytics platform** consolidating supply, demand, inventory, and planning reports into a single actionable view.

Contact : Zheng Li, Maria Chevas, Balaji Nagaraj - zheng_li@amat.com,balaji_nagarajan@amat.com;

Senior Business Intelligence Analyst

Applied Materials Singapore & India

12/2013 - 07/2019

Achievements/Tasks

- Executed the Business Intelligence Stack for integrating Applied Global service business Unit information systems into a 'Single Source of Truth', for enabling faster, flexible reporting on business volume, opportunity size, pricing, potential for market share regain, product performance and usage-per-customer
- Enabled planners to **monitor fill rate, supplier reliability, and backlog trends in near real time**, improving visibility across the end-to-end supply planning process.
- Automated data ingestion and reporting workflows, **eliminating manual spreadsheet-based tracking and reducing planning cycle time by 50–60%** .
- Delivered **exception-based alerts and prioritized insights**, allowing planners to focus on high-risk materials and suppliers, improving response time and decision quality.
- Developed and maintained a suite of data visualizations to support data-driven decision making, resulting in a 20% increase in data accessibility and a 15% increase in data-driven decision making across the organization.
- Used a variety of machine learning algorithms, including supervised learning (linear regression, logistic regression, decision trees, random forests, gradient boosting machines), unsupervised learning, natural language processing (NLP), and deep learning. Worked with large datasets (millions of rows and hundreds of columns) using big data technologies such as Hadoop, Spark, and Hive.
- Managing the business intelligence, analytic & operational reporting required for achieving customer service. Bringing intelligent capabilities & frameworks in data platform to improve productivity and efficiency of all persona's objectives & other needs across Organization

Contact : Cynthia Brown, Balaji Nagarajan, Gopala ManjeGowda, Prakash Krishnaswamy - balaji_nagarajan@amat.com;gopala_manjegowda@amat.com



CERTIFICATES

Data Science in Python

University of Michigan

Tableau Desktop Specialist Certification

Data Cleaning in Python

Engineering for Generative AI

Claud 101 (02/2026 - Present)

Anthropic.skilljar